

# APOORV DIXIT

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## PROJECTS

- **LLM Inference Optimization** - Built an LLM inference benchmarking framework for GPT-2 (124M), comparing PyTorch FP32/FP16, **SDPA**, torch.compile, **ONNX** Runtime, and **TensorRT** across batch sizes 1–16 and sequence lengths 64–256; measured latency, throughput, and GPU memory, achieving **~3.4× higher throughput** (110K vs 32K tokens/s) and **~46% lower latency** (16.5 ms vs 30.3 ms) with TensorRT over ONNX.
- **LLM Serving Infrastructure** - Built a production-style GPU inference serving platform benchmarking HuggingFace **FastAPI**, **vLLM**, **SGLang**, **TensorRT-LLM**, and NVIDIA **Triton** for GPT-2 FP16 workloads on an RTX 4080 Laptop GPU; measured p50/p95 latency, throughput, and GPU utilization, with SGLang achieving **3.2× lower latency** (80 vs 257 ms) and **3.2× higher throughput** (12.4 vs 3.9 req/s) than HuggingFace and Triton baselines while TensorRT-LLM delivered comparable performance through optimized LLM inference.
- **LLM Quantization Benchmarking** - Benchmarked GPT-2 across 5 precision modes (FP32, FP16, INT8, INT4, **NVFP4**) using PyTorch, optimum-quanto, and torchao; FP16 delivered 3.2× lower latency, 120K tokens/s throughput, and 47% memory reduction with negligible accuracy loss; validated quantization tradeoffs via perplexity-based evaluation across all precision levels
- **Foundation Model Quantization & Optimization** - Quantized Lag-Llama time-series foundation model using 4-bit **bitsandbytes** NF4 and PyTorch **torchao** INT4 weight-only quantization; benchmarked latency, memory usage, and zero-shot forecasting accuracy across multiple datasets, achieving up to 39% GPU memory reduction while analyzing quantization kernel constraints and accuracy-performance tradeoffs
- **Distributed Deep Learning Training Systems** - Implemented BERT-base (110M) distributed fine-tuning on GLUE SST-2 across baseline PyTorch, manual DDP, production **DistributedDataParallel**, and a from-scratch multiprocessing-based DDP implementation; rebuilt gradient synchronization using custom broadcast/all-reduce primitives to demonstrate model replication, distributed sampling, and process coordination
- **AI Workload Analysis** - Conducted comparative performance analysis of **GPT-2**, **DistilGPT-2**, **BERT-base**, and **ResNet-50**, measuring latency, throughput, memory footprint, and FLOPs; identified batching saturation at batch sizes 8–16 and achieved up to 39% MFU during FP16 inference
- **GPU Profiling & Analysis** - Analyzed GPT-2 inference performance with PyTorch Profiler and NVIDIA Nsight Systems; identified GPU bottlenecks, optimized batch scaling, and achieved ~105K tokens/sec throughput.

## EXPERIENCE

### JPMorgan Chase & Co.

Jun 2025 - Present | Jersey City, NJ

Quantitative Analytics Associate

- Architected and productionized a GPT-5 (**AzureOpenAI**) + Ensemble **RAG** system (**LangChain**, Chroma Vector DB, BM25, Parent Document Retriever) to auto-generate structured risk review drafts from 80-400 page model documents, reducing enterprise model review cycle time by **33%** (3 months to 2 months) across **500+** analysts.
- Implemented a **TabTransformer** in PyTorch on 300K loan records (11.5:1 imbalance), benchmarking against **Logistic Regression** (0.58 ROC-AUC) and **XGBoost** (0.65), achieving 0.61 ROC-AUC while evaluating attention-based feature interaction learning for credit risk modeling
- Conducted assessment of a **Regression-based** Auto loans recovery forecasting model (20K records) supporting an \$85B auto-loans portfolio, executing marginal vs conditional **R<sup>2</sup>** analysis, **KS/CSI** stability tests, **VIF** multicollinearity checks, **heteroskedasticity** diagnostics, and backtesting to surface structural risks prior to CCAR-aligned regulatory review
- Reconstructed and audited an **XGBoost** regression pipeline (20K datapoints) predicting 5 venture finance metrics using log transforms, k-fold CV, Random Search hyperparameter tuning, Recursive Feature Elimination, and SHAP explainability, identifying 7 risks before publication

### JPMorgan Chase & Co.

Jun 2024 - Aug 2024 | Jersey City, NJ

Quantitative Analyst Intern

- Built an LLM-based benchmarking platform evaluating 8 document extraction models, achieving 94% F1 on CoNLL-2003
- Developed a **OpenCV** based document stress-testing engine generating 10+ noisy variants per file to evaluate robustness of 8 extraction models

### Siemens Digital Industries Software

Jul 2021 - Jul 2023 | Pune, India

Associate Software Engineer

- Designed a two-stage AI pipeline (**OpenCV + Tesseract/Google Vision OCR + SpaCy/BERT NER**) to convert scanned PDF manufacturing drawings into SolidEdge-native 2D CAD files by extracting geometric primitives (73% shape accuracy) and metadata tables (90% NER accuracy), integrating outputs via SolidEdge APIs to prototype a **top customer-requested** automation feature in the CAD modeling domain
- Developed an early **GPT-3.5-powered conversational assistant** integrated into a web-based CAD editor (Zel-X) that leveraged help documentation to guide users through complex modeling workflows, prototyping an LLM-driven support interface to improve user experience
- Delivered 9 features and resolved 50+ issues across two cloud SaaS products while increasing automated test coverage to 98%.

## SKILLS

Python, PyTorch, Hugging Face Transformers, ONNX Runtime, TensorRT, TensorRT-LLM, vLLM, SGLang, NVIDIA Triton, torch.compile, torch.distributed, DDP, NVIDIA Nsight Systems, Nsight Compute, PyTorch Profiler, CUDA Events, FP16/INT8/INT4 Quantization, bitsandbytes, torchao, FastAPI, Azure OpenAI, LangChain, RAG, XGBoost, TabTransformer, SHAP, OpenCV, Pandas, NumPy, C++, CUDA C++

## EDUCATION

University of Southern California

Aug 2023 - May 2025 | Los Angeles, USA

Master of Science in Computer Science

GPA - 3.73 / 4

Pune Institute of Computer Technology

Aug 2017 - April 2021 | Pune, India

Bachelor of Engineering in Computer Engineering

GPA - 9.14 / 10